SOMAIYA VIDYAVIHAR UNIVERSITY
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(Somaiya Vidyavihar University)

Department of Computer Engineering

Somaiya TRUST

Course Name:	Web Development Laboratory	**Semester:**	**III**
Date of Performance:	___ / ___ / _____	**DIV/ Batch No:**	
Student Name:		**Roll No:**	

Experiment No: 1

Title

Design and Development of a Heatwave Intelligence Web Portal Front-End using HTML5

Aim of the Experiment

To design and develop a multi-page, use-case-based web portal front-end using HTML5 for a Climate Intelligence for Heatwave Monitoring, Prediction, and Early Warning system (KJS-CES-01), demonstrating proper document structure, semantic elements, formatting, lists, links, images, image maps, tables, forms, and media elements.

Objectives of the Experiment

1. Develop structured and semantic HTML5 documents for a real-world climate-intelligence dashboard.
2. Create interactive and media-rich web pages for heatwave monitoring and early warning.
3. Apply a real-world AI use case (heatwave monitoring) as the context for front-end web development.
4. Build a basic but complete front-end web portal that serves as the foundation for later CSS3, JavaScript, and React work.

COs to be achieved

CO1: Developing webpages using HTML and CSS.

Books / Journals / Websites references

1. *(Students should write — e.g., MDN Web Docs HTML5 reference, W3C.)*

Theory

HTML5 is the standard markup language used to structure content on the web. It introduces semantic elements such as <header>, <nav>, <main>, <section>, <article>, <aside>, and <footer> that describe the meaning of content rather than only its appearance. HTML5 also

adds native support for media (<audio>, <video>), graphics (<canvas>, SVG), richer form controls (e.g., email, date, number, range input types with built-in validation), and embedding via <iframe>. These features make it well suited to building a decision-support dashboard such as a Heatwave Intelligence portal, where structured data (region-wise temperatures, forecasts, alerts) must be presented clearly and accessibly to meteorologists, disaster-management officers, and citizens.

Problem Statement

"Climate Intelligence Heatwave Monitoring & Early-Warning Web Portal"

A meteorological/disaster-management agency wants to develop a basic HTML5-based web portal front-end to display region-wise temperature data, heatwave forecasts, hotspot maps, alerts, and advisory services for meteorologists, disaster-management officers, and the public. Each task below contributes one page or feature of this portal, mirroring the components of the Heatwave Intelligence dashboard.

Task 1: Home Page – HTML5 Document Structure & Global Attributes

Use Case: Design the Home Page of the Heatwave Intelligence Portal.

Requirements:

- Use proper HTML5 document structure.
- Display the system name ("Heatwave Intelligence System"), a logo text, and the current monitoring season/year.
- Use global attributes (id, class, title, style).
- Add a tooltip on the system name explaining its purpose.

Deliverable: index.html

Skills Tested: Document structure, global attributes.

Task 2: Alert Description Page – Formatting & Text-Level Tags

Use Case: Create a page describing a Severe Heatwave Alert Bulletin.

Requirements:

- Highlight the alert name/region using and <mark>.
- Use for advisory themes and <small> for disclaimers (e.g., "forecast subject to revision").
- Use <sup> and <sub> for units and dates (e.g., temperature 47°C, bulletin version 2.1).
- Use <abbr> for technical terms (e.g., AWS, IMD, HW, SHW).

Deliverable: alert-details.html

Skills Tested: Formatting & text-level tags.

Task 3: About System Page – Block vs Inline Formatting

Use Case: Design an About the Heatwave Intelligence System page.

Requirements:

- Use sections for Objective, Data Sources, and Stakeholders.
- Use <blockquote> for a message from the Chief Meteorologist / Disaster Management Officer.
- Apply inline formatting inside paragraphs.
- Apply background colours to block sections (e.g., severity colour coding).

Deliverable: about-system.html

Skills Tested: Block-level and inline elements.

Task 4: Data Sources & Stakeholders Page – Lists & Nested Lists

Use Case: Display the Data Sources and Stakeholder Roles.

Requirements:

- Ordered list for the forecast/alert workflow steps.
- Unordered list for data sources (AWS sensors, IMD database, satellite feeds).
- Nested list showing stakeholders and the actions available under each role.
- Description list (<dl>) explaining abbreviations (AWS, IMD, HW, SHW).

Deliverable: sources.html

Skills Tested: List tags.

Task 5: Navigation Page – Hyperlinks

Use Case: Create a Navigation Menu for the portal.

Requirements:

- Link to all internal pages created so far.
- External link to the India Meteorological Department (IMD) website.
- Email link to the heatwave advisory coordinator.
- Telephone link for the disaster-management helpline.

Deliverable: navigation.html

Skills Tested: Hyperlink tags.

Task 6: Hotspot Map – Image & Image Map

Use Case: Provide an interactive Regional Heatwave Hotspot Map.

Requirements:

- Display a regional map image.
- Create an image map with clickable areas:
  - Region A (e.g., North Zone)
  - Region B (e.g., Central Zone)
- Each area opens a relevant region-detail page or link.

Deliverable: hotspot-map.html

Skills Tested: Image and image maps.

Task 7: Forecast Summary Page – Tables

Use Case: Display Region-wise Temperature & Forecast Results.

Requirements:

- Table showing Region, Station, Max Temperature, Severity Level.
- Use <caption>, <thead>, <tbody>, <tfoot>.
- Merge header cells using colspan (e.g., a "Forecast" header spanning multiple columns).

Deliverable: forecast.html

Skills Tested: Table tags.

Task 8: Advisory Subscription – Forms

Use Case: Create an Online Heatwave Advisory Subscription Form.

Requirements:

- Citizen name, email, mobile number.
- Region selection (dropdown).
- Preferred alert language (radio).
- Alert types to receive (checkbox).
- Submit and reset buttons.

Deliverable: subscribe.html

Skills Tested: Form tags.

Task 9: Media & Frames – Executable Content

Use Case: Create a Media & Awareness Page.

Requirements:

- Embed a previous heatwave awareness video using <video>.
- Add background audio advisory using <audio>.
- Embed a Google Map / IMD site using <iframe>.
- Use a <noscript> message for users with JavaScript disabled.

Deliverable: media.html

Skills Tested: Frames & executable tags.

Code :

Output / Screenshot:

Conclusion and Discussion:
(students should write in your words) (Also take feedback from fellow classmates and faculty to add here)